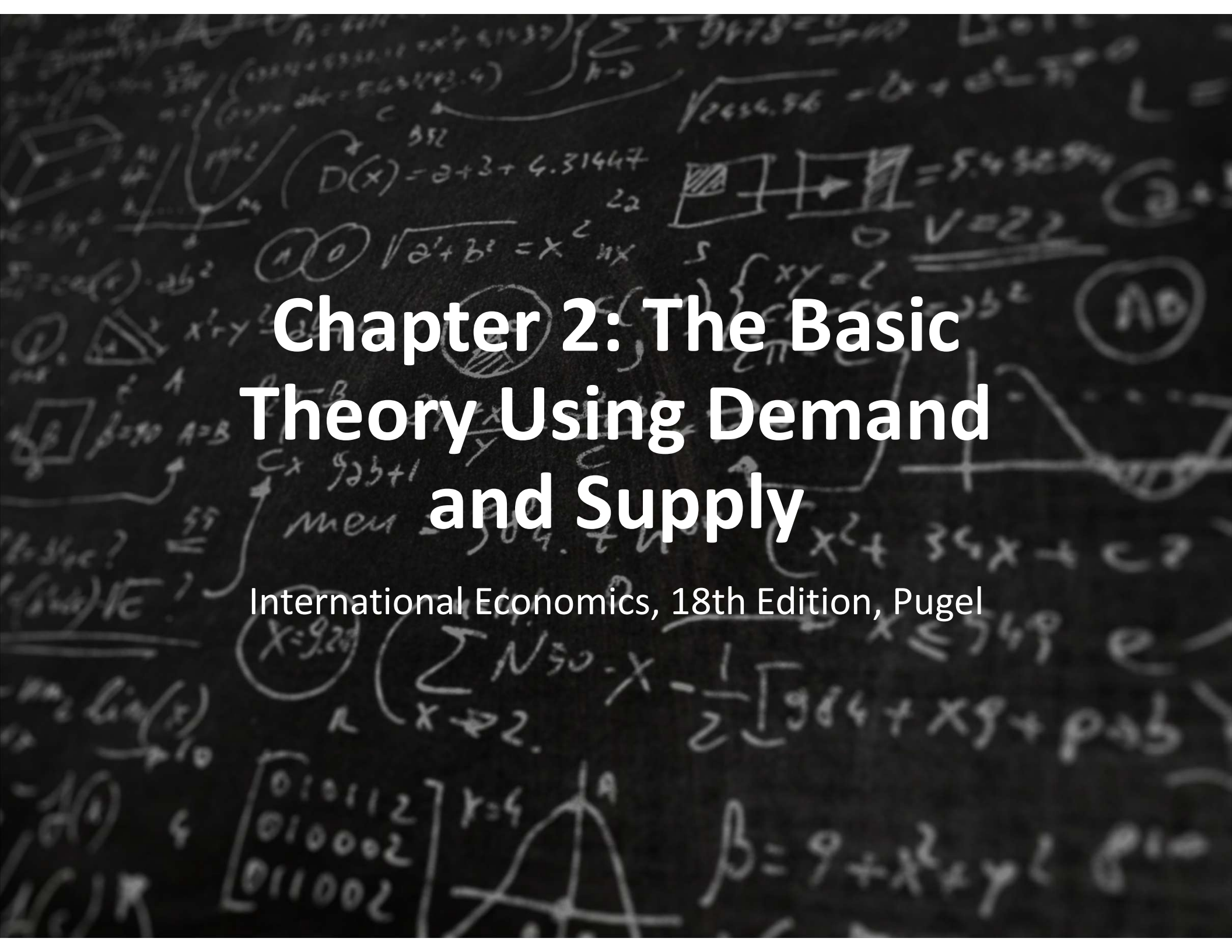




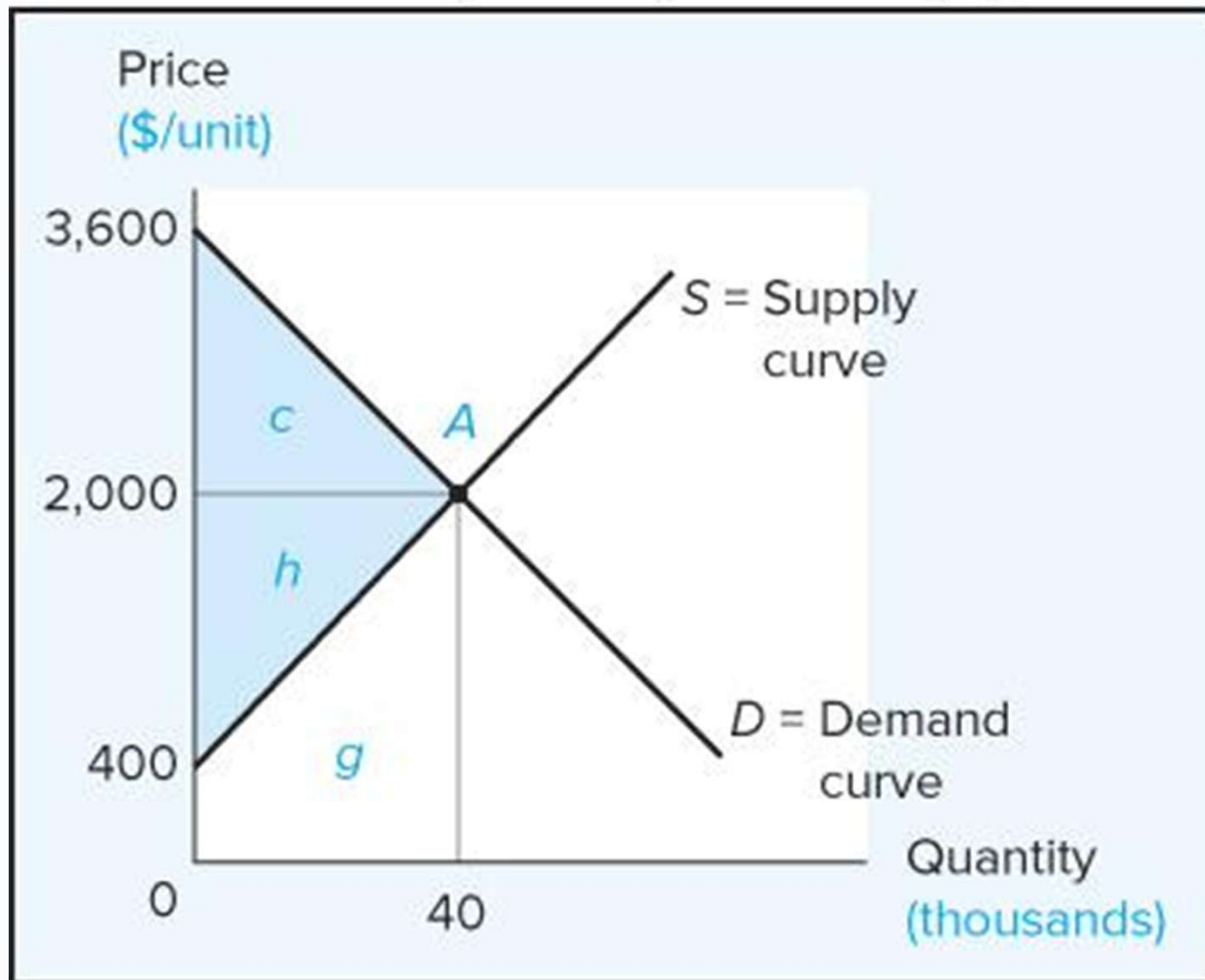
The background is a dark, textured surface resembling a chalkboard, covered with various handwritten mathematical expressions and diagrams in white. These include algebraic equations like $D(x) = 2 + 3 + 4.31447$, $\sqrt{a^2 + b^2} = x^2$, and $\beta = 9 + x^2 + y^2$; geometric diagrams such as rectangles and triangles; and other notations like $\sum$, π , and Λ . The text is centered over this background.

Chapter 2: The Basic Theory Using Demand and Supply

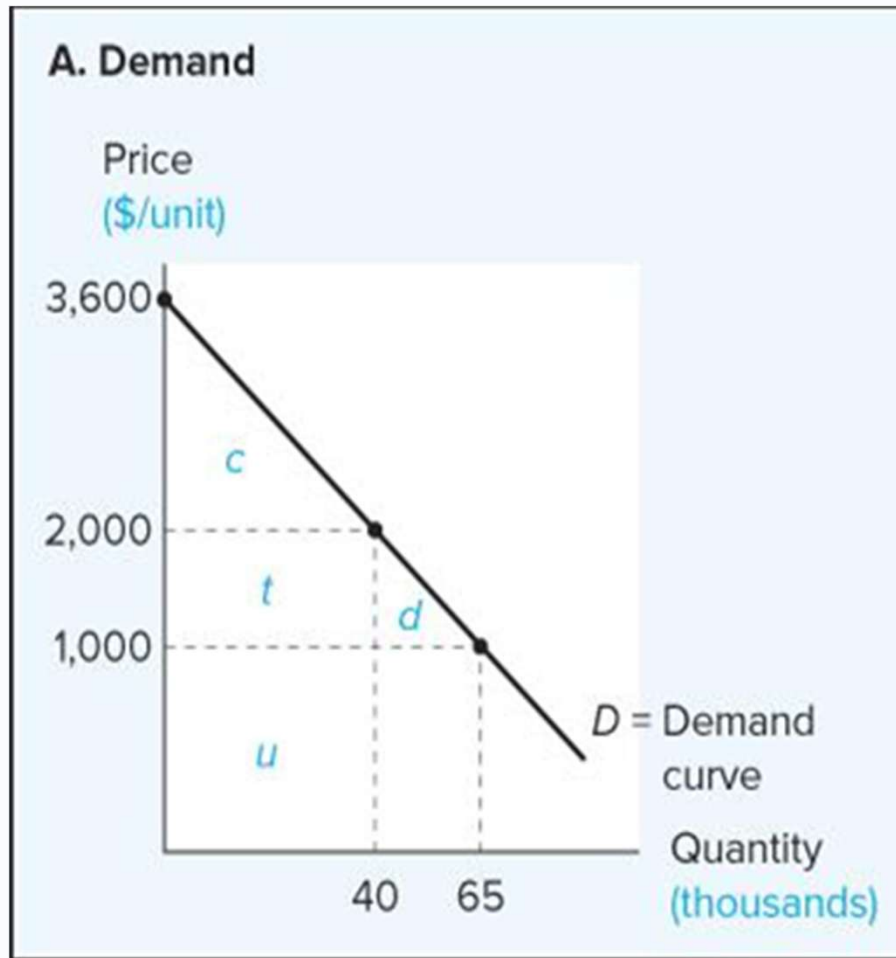
International Economics, 18th Edition, Pugel

The Market for Motorbikes: Demand and Supply

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Demand Curve



Shows the quantity consumers will buy at each possible market price, other things equal.

$$P = a - bQ_d$$

$$a = 3,600$$

$$b = -\frac{1,000}{65k - 40k} Q_d$$

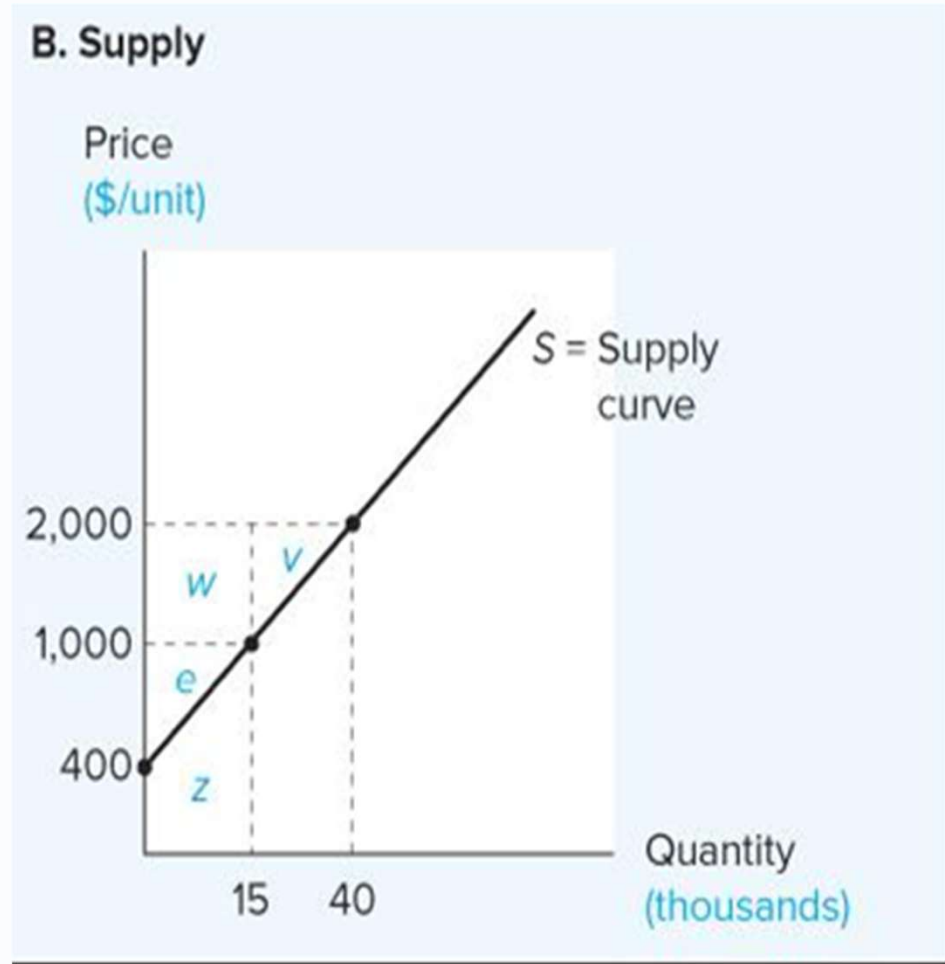
$$P = 3600 - 0.04 Q_d$$

Supply Curve

Shows the quantity that producers will offer for sale at each possible market price, other things equal.

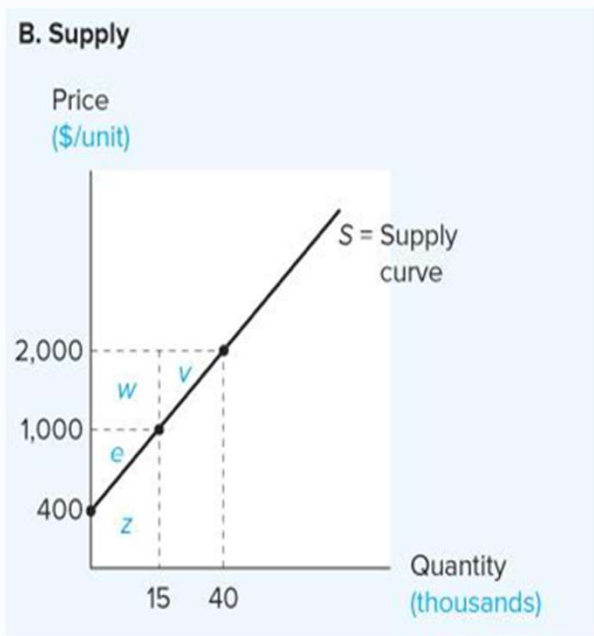
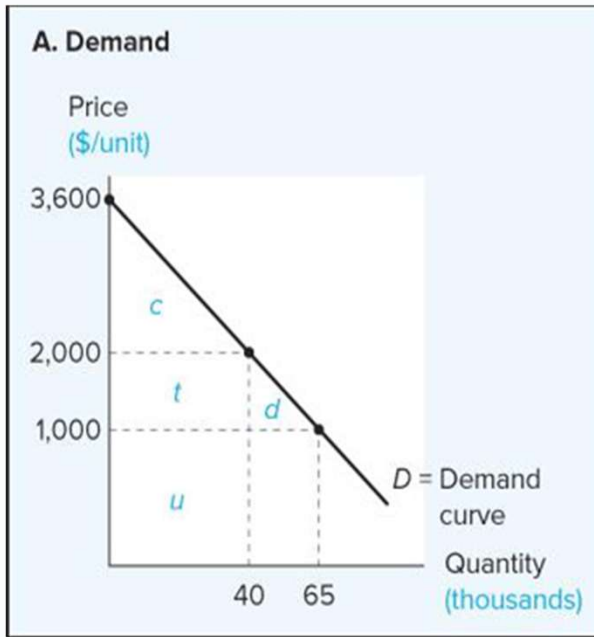
$$P = a + bQ_s$$
$$a = 400$$
$$b = \frac{1000}{(40k - 15k)}$$

$$P = 400 + 0.04 Q_s$$



Simultaneous Equations

Calculate the intersection of Demand & Supply



$$P = 3600 - 0.04 Q_d \quad \& \quad P = 400 + 0.04 Q_s$$

$$3600 - 0.04 Q_d = 400 + 0.04 Q_s$$

$$3600 - 0.04 Q \quad - 400 = 400 + 0.04 Q \quad - 400$$

$$3200 - 0.04 Q \quad + 0.04 Q = 0.04 Q \quad + 0.04 Q$$

$$\frac{3200}{0.08} = \frac{0.08 Q}{0.08}$$

$$Q = 40,000$$

$$P = 3600 - 0.04 \times 40,000 = \$2,000$$

Supply & Demand Elasticity

Elasticity is a measure of responsiveness that is “unit-free”

Price elasticity of demand

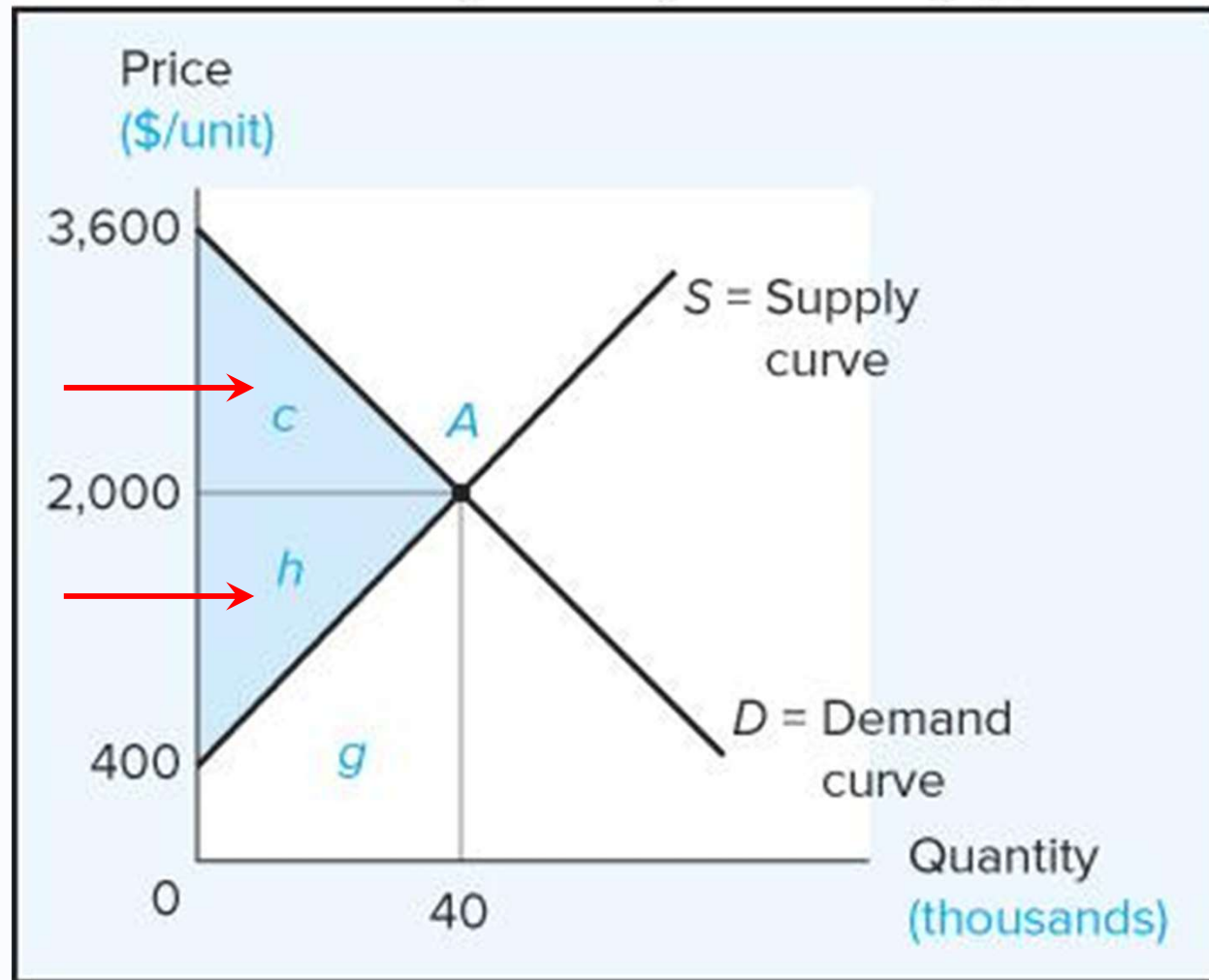
% change in quantity demanded resulting from a 1% price change.

Price elasticity of supply

% increase in quantity supplied resulting from a 1% increase in price.

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Two National Markets and the Opening of Trade

Arbitrage is buying something in one market and reselling the same thing in another market to profit from a price difference.

With no trade, differences in the price of a product between national markets creates an arbitrage opportunity that can lead to international trade.

Retail Arbitrage



Part 2: Free Trade Equilibrium

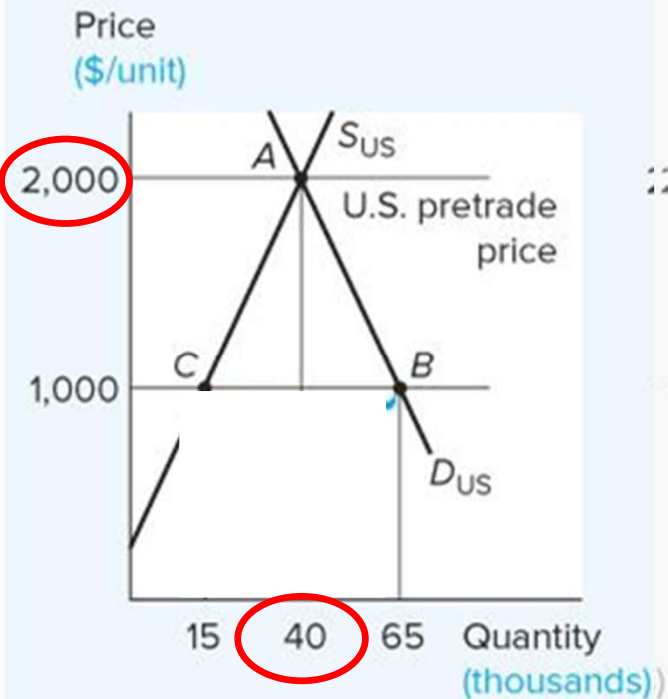
Free Trade Equilibrium

As international trade in a product (e.g., motorbikes) develops between the two countries, it affects the equilibrium price in importing and exporting countries.

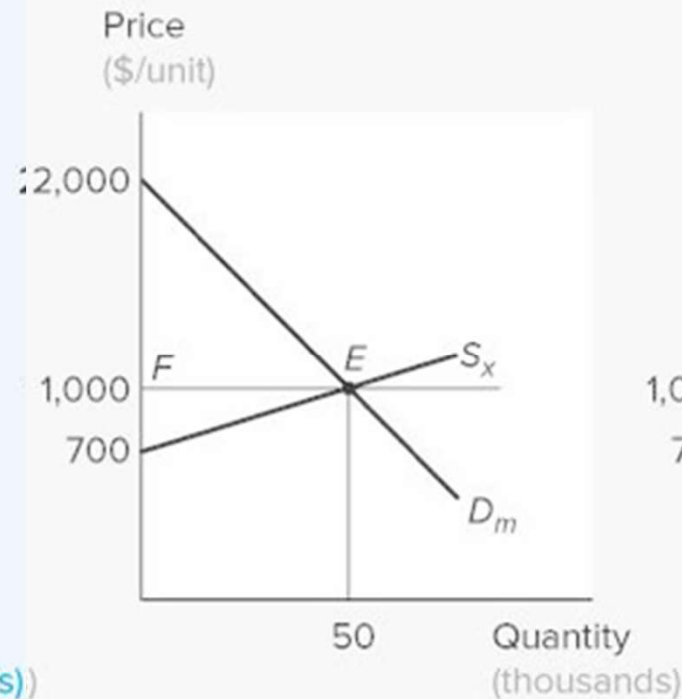
The Effects of Trade on Production, Consumption, and Price with Demand and Supply Curves

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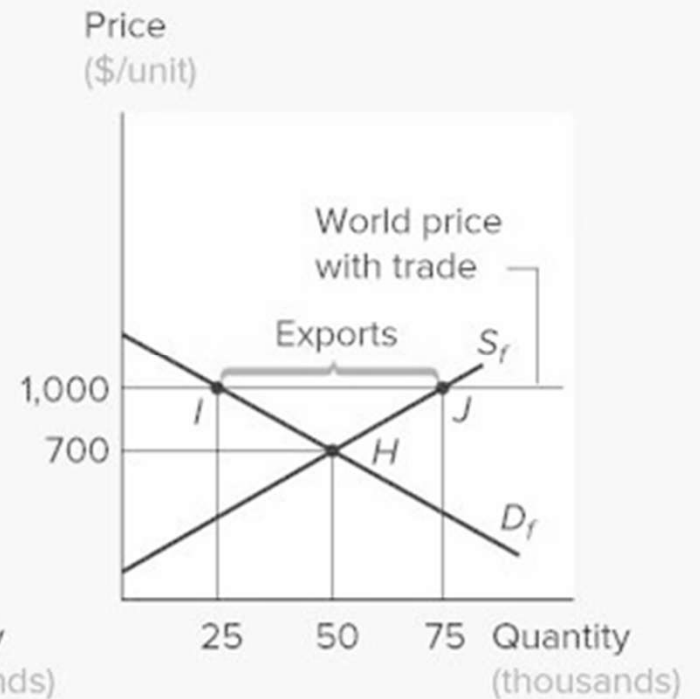
A. The U.S. Motorbike Market



B. International Motorbike Market

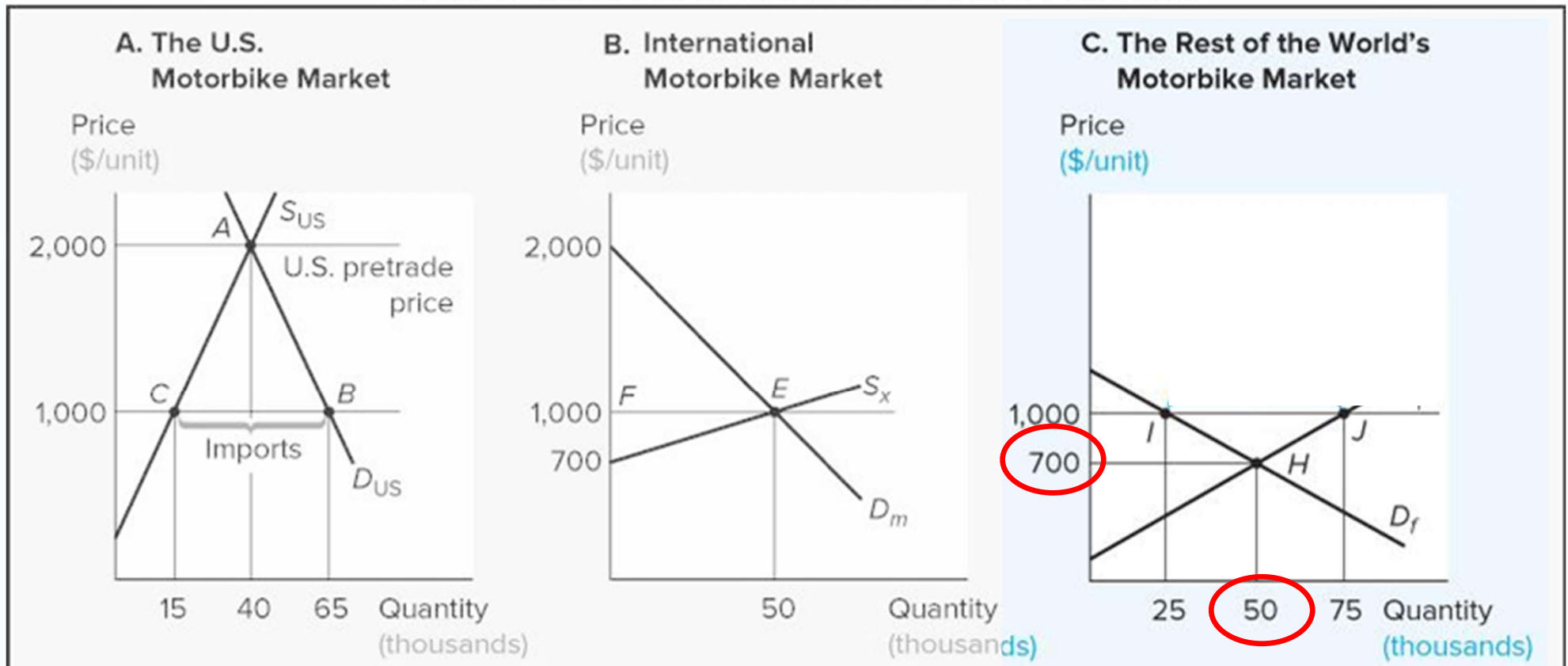


C. The Rest of the World's Motorbike Market



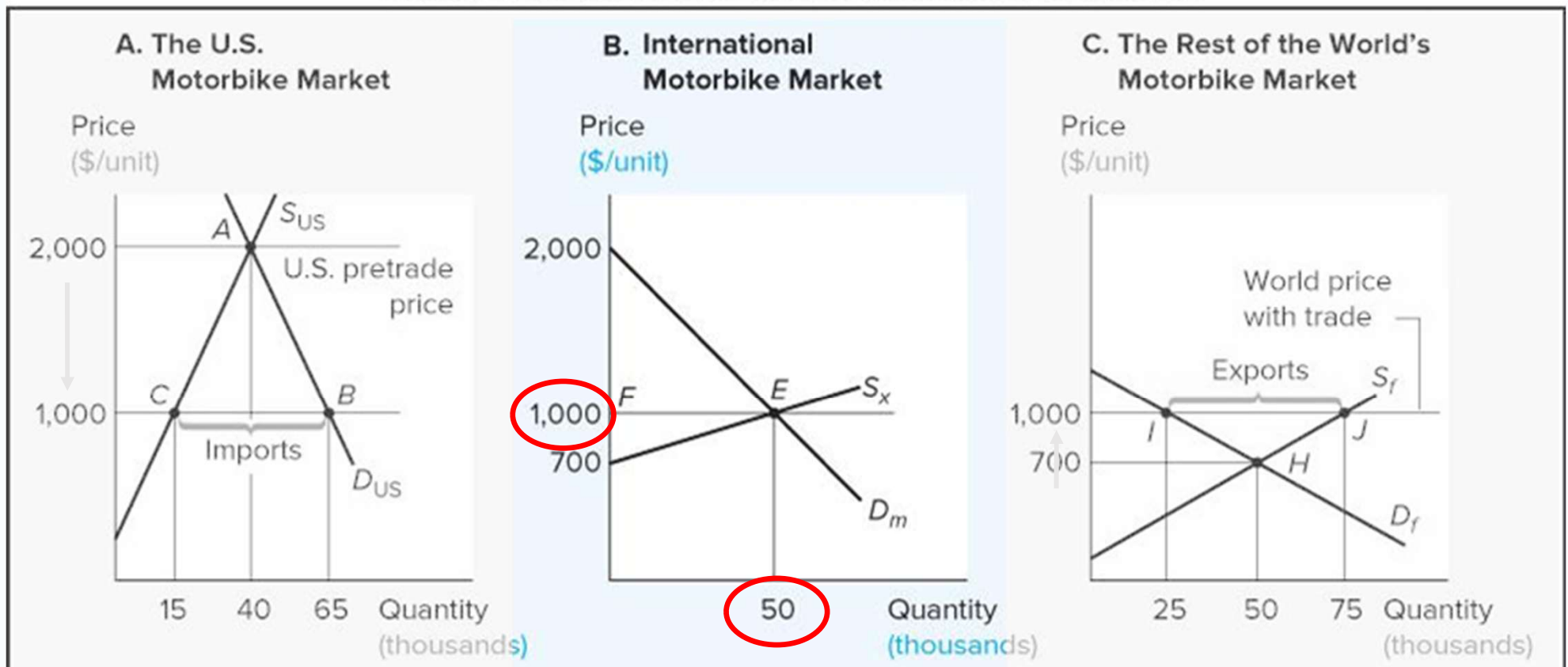
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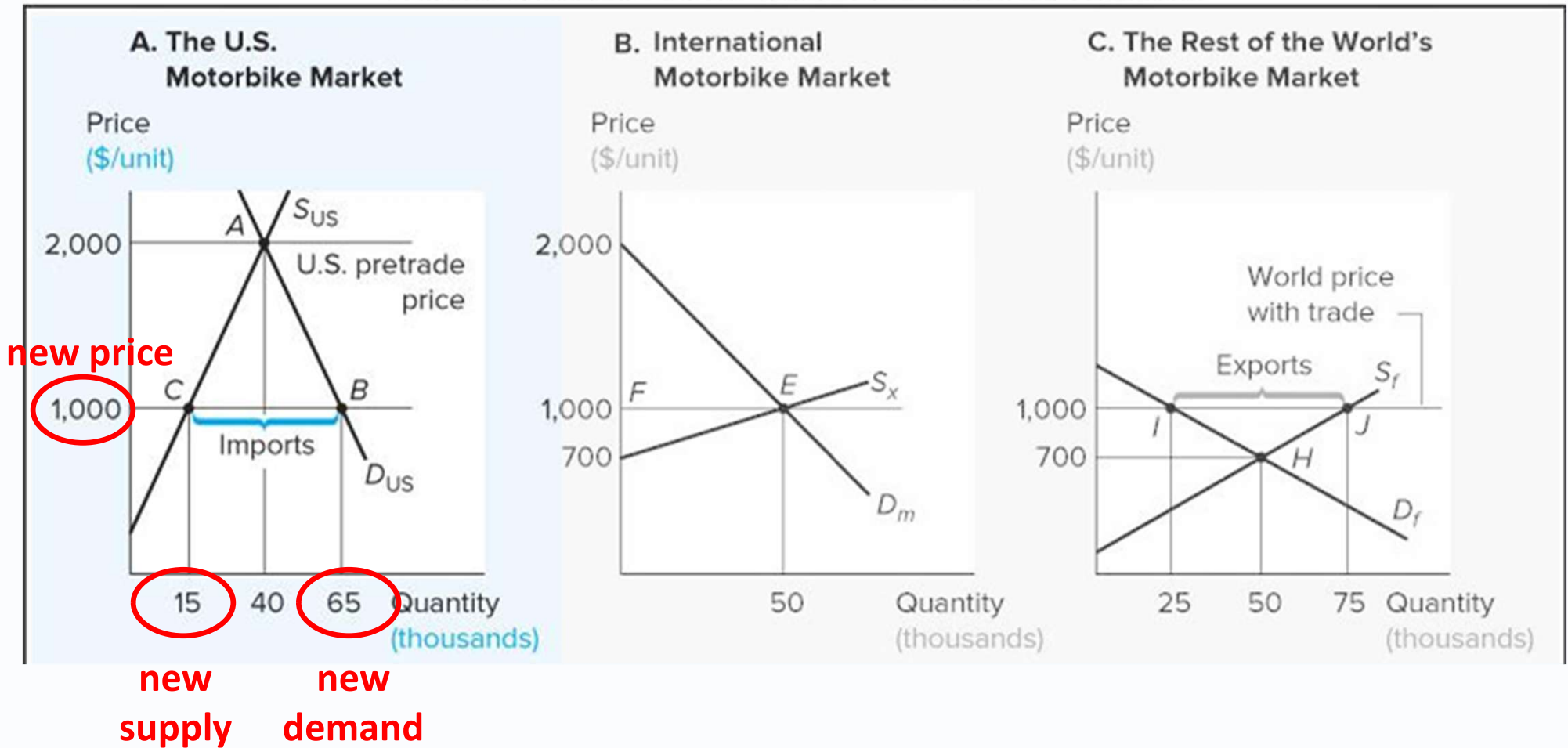
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excess
demand

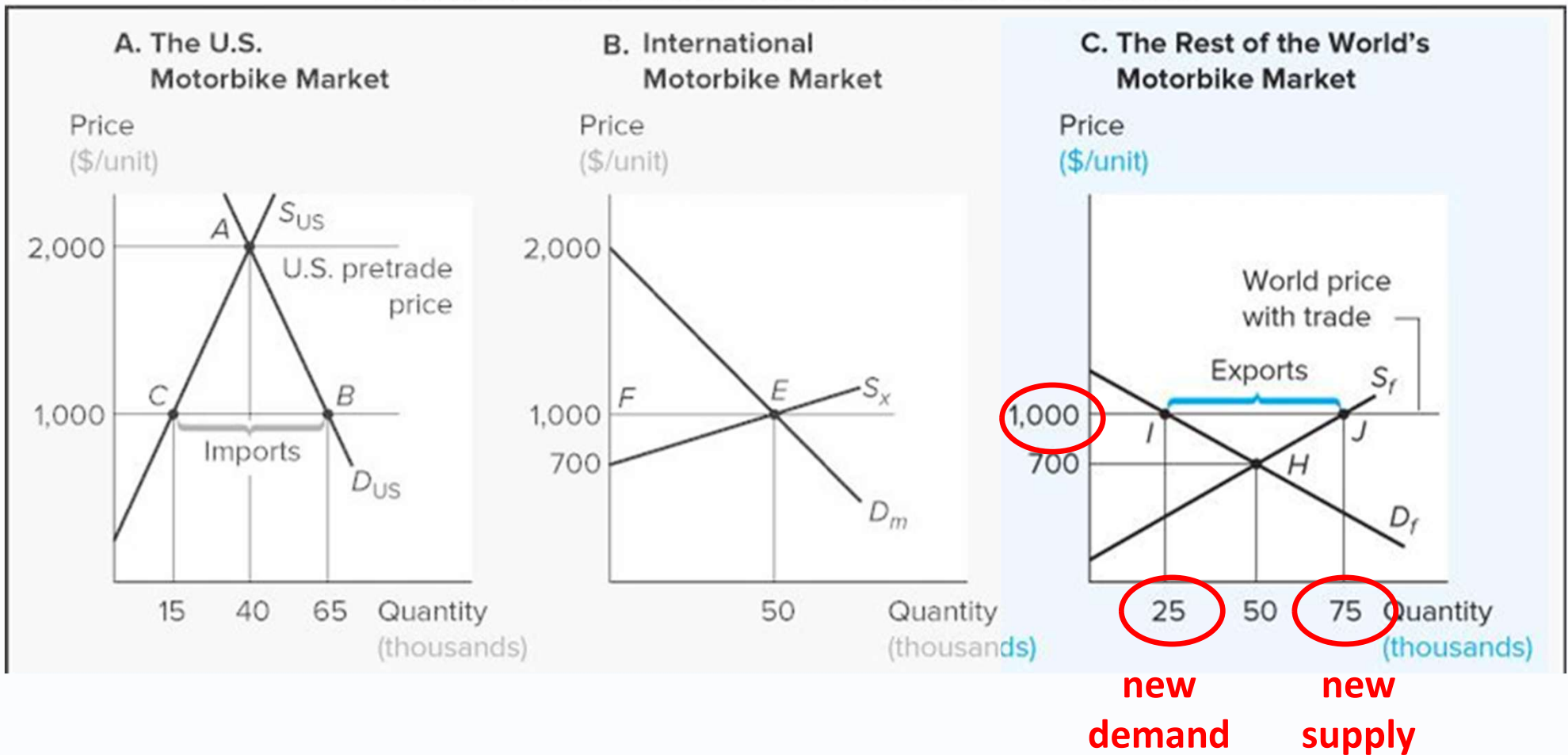
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The Effects of Trade on Production, Consumption, and Price with Demand and Supply Curves

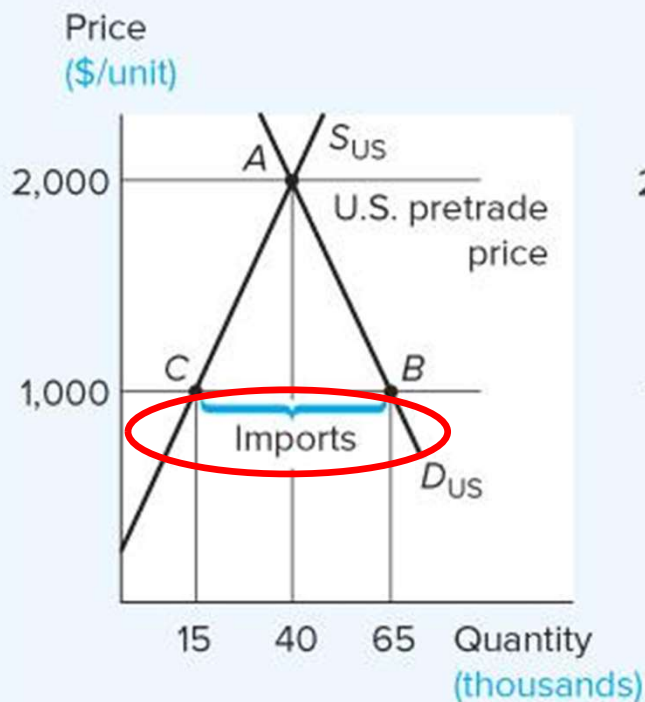
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The Effects of Trade on Production, Consumption, and Price with Demand and Supply Curves

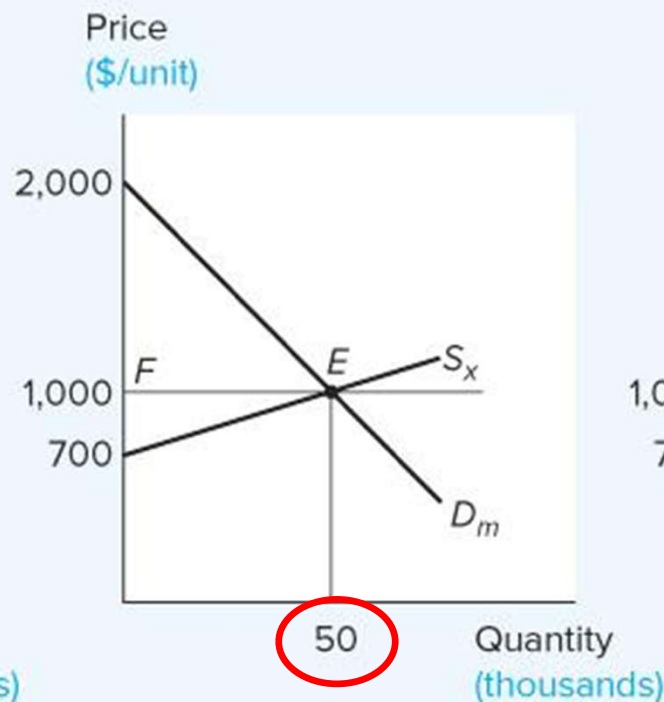
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A. The U.S. Motorbike Market



S_{US} = U.S. supply
 D_{US} = U.S. demand

B. International Motorbike Market



S_x = Rest-of-world supply of exports
 $(S_x = S_f - D_f)$
 D_m = U.S. demand for imports
 $(D_m = D_{US} - S_{US})$

C. The Rest of the World's Motorbike Market



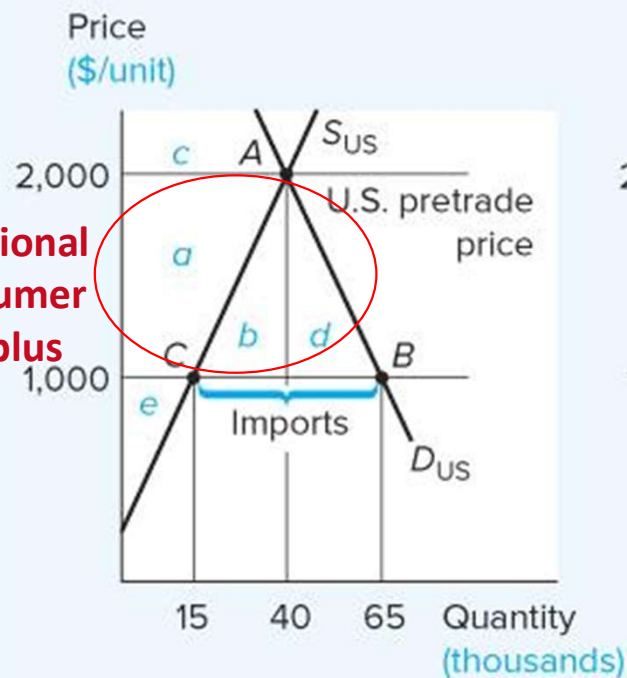
S_f = Rest of world's supply
 D_f = Rest of world's demand

The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

Additional consumer surplus = $a + b + d$

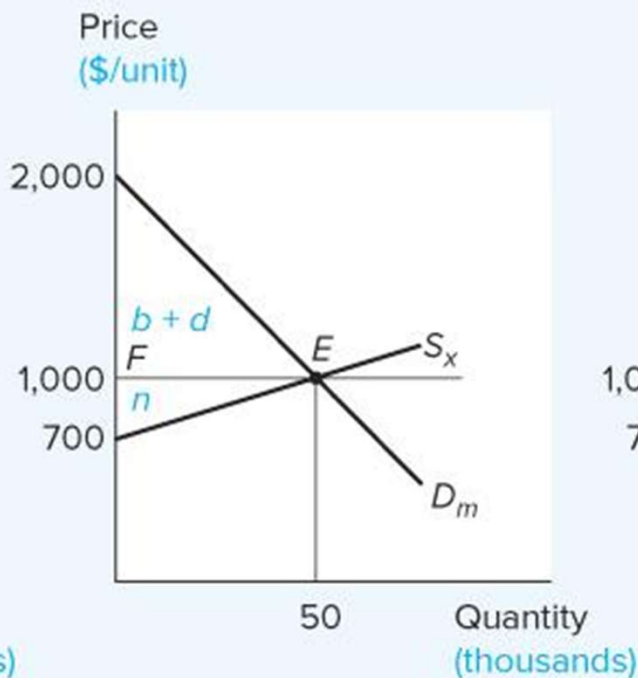
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A. The U.S. Motorbike Market



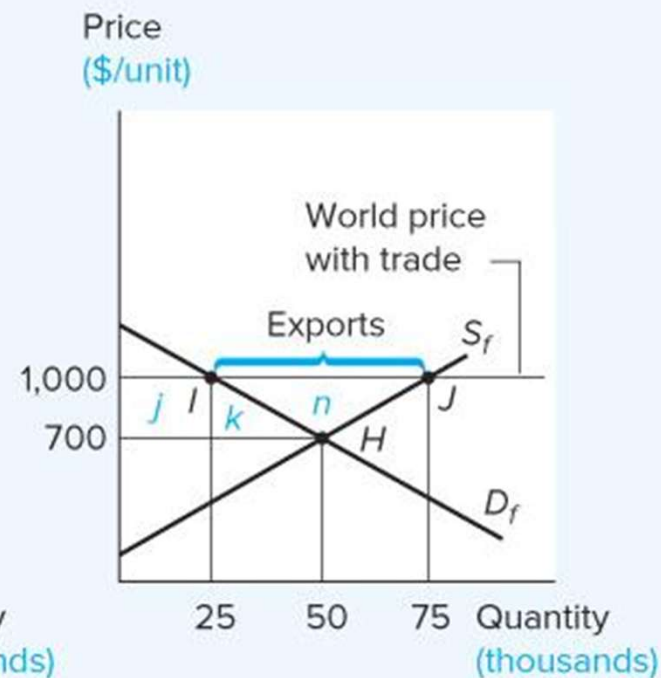
S_{US} = U.S. supply
 D_{US} = U.S. demand

B. International Motorbike Market



S_x = Rest-of-world supply of exports
 D_m = U.S. demand for imports

C. The Rest of the World's Motorbike Market



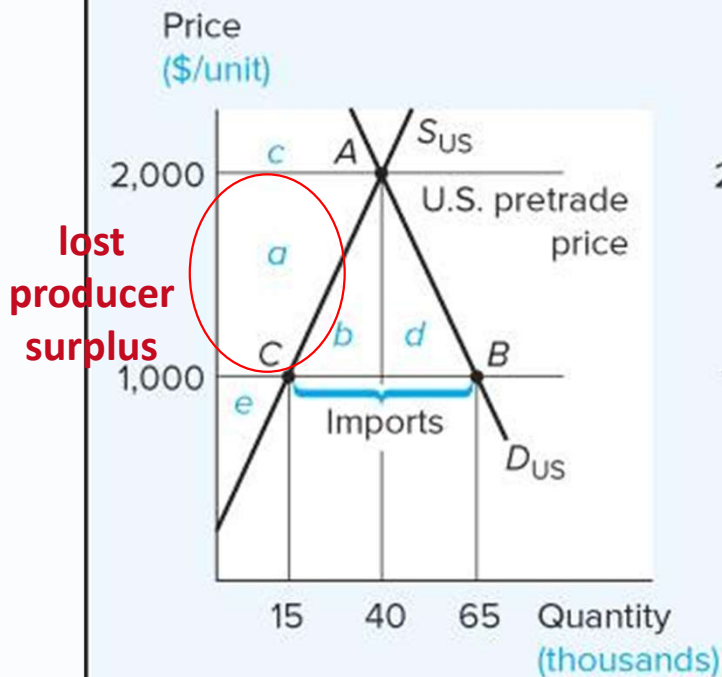
S_r = Rest of world's supply
 D_r = Rest of world's demand

The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

Lost producer surplus = -a

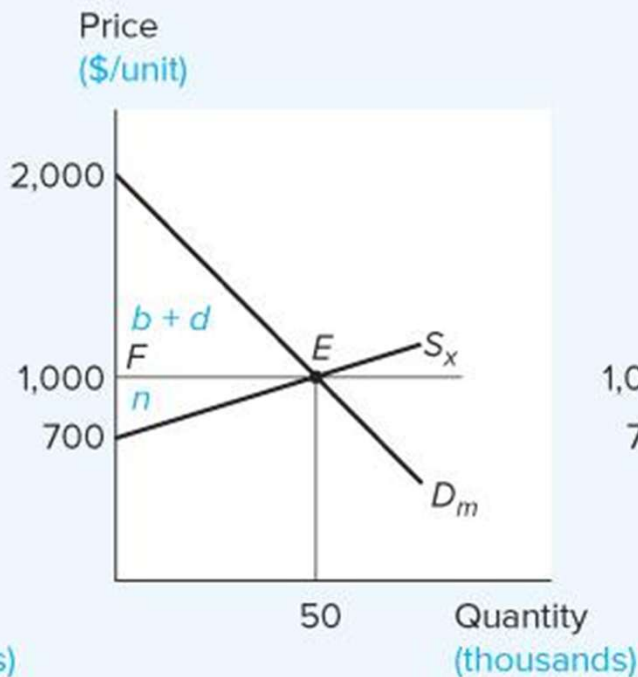
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A. The U.S. Motorbike Market



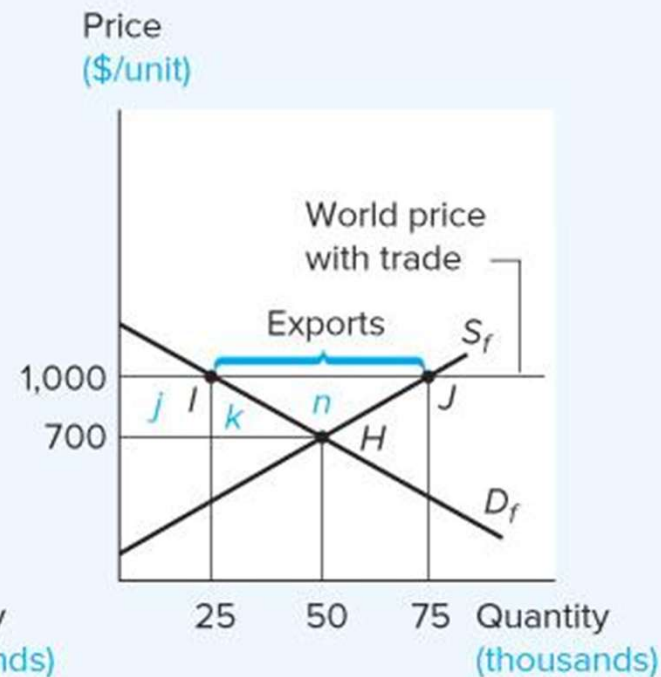
S_{US} = U.S. supply
 D_{US} = U.S. demand

B. International Motorbike Market



S_x = Rest-of-world supply of exports
 D_m = U.S. demand for imports

C. The Rest of the World's Motorbike Market

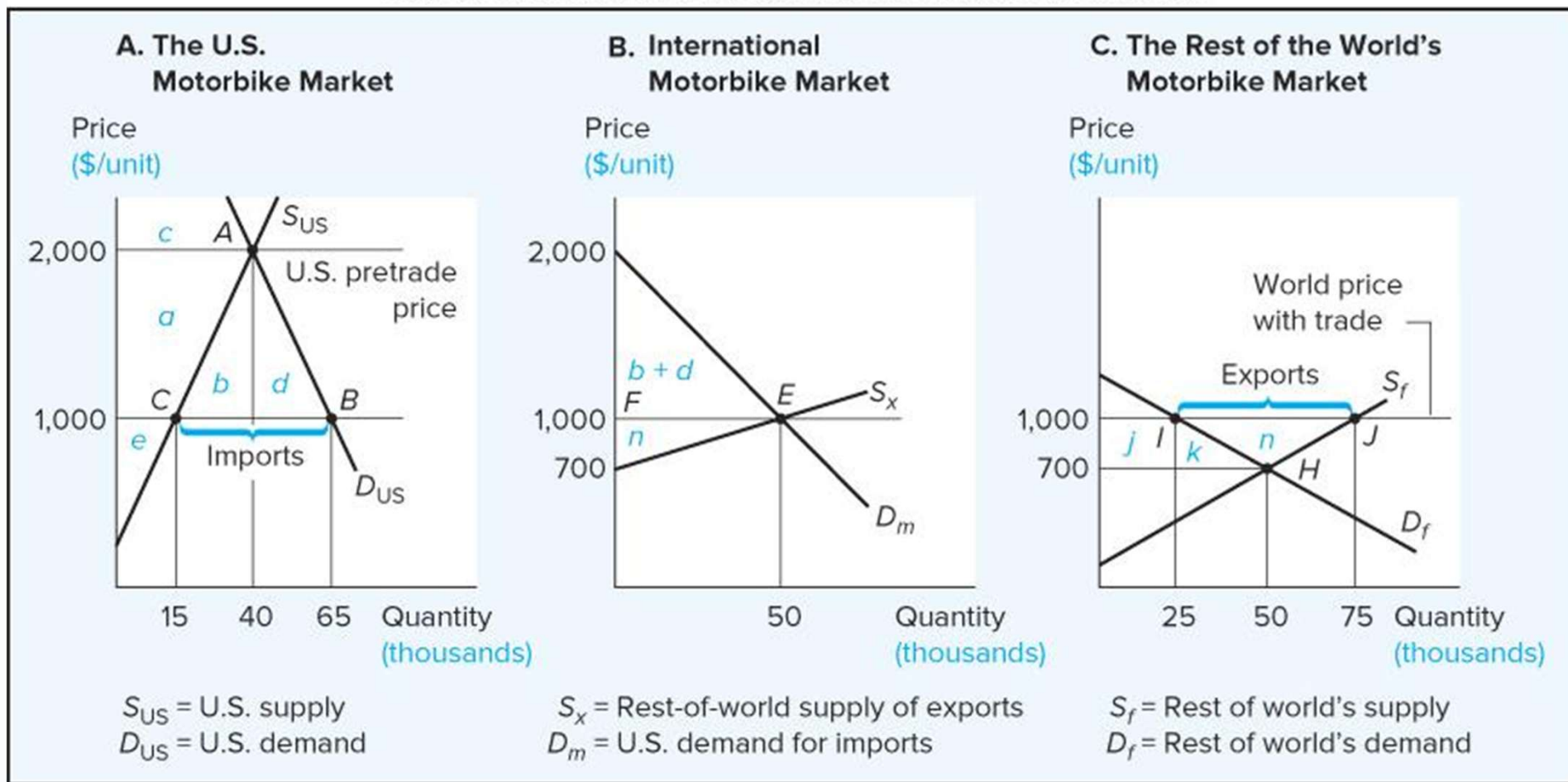


S_r = Rest of world's supply
 D_r = Rest of world's demand

The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

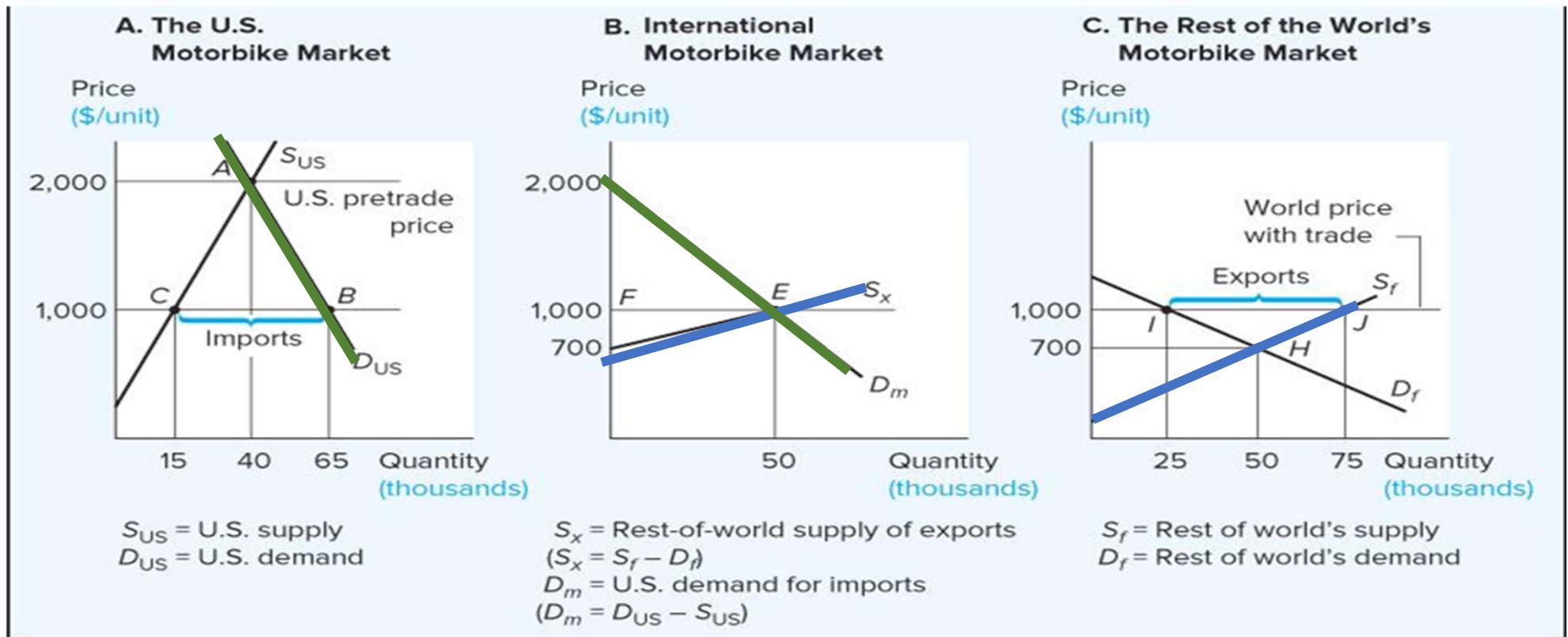
$$\begin{aligned} \text{Net US gain/loss} &= \text{consumer surplus} - \text{producer surplus} \\ &= (a + b + d) - a = \underline{b + d} \end{aligned}$$

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Which Country Gains More?

The side with the less elastic (steeper) trade curve (import demand curve or export supply curve) gains more.



Summary: Welfare Effects of Free Trade

United States (Importing Country)

Group	Surplus with Free Trade	Surplus with No Trade	Net Effect of Trade
Consumers	$a + b + c + d$	c	$a + b + d$
Producers	e	$a + e$	$-a$ [a loss]
U.S. as a whole (consumers + producers)	$a + b + c + d + e$	$c + a + e$	$b + d$

Rest of the World

Group	Net Effect of Trade
Consumers	$-(j + k)$ [a loss]
Producers	$j + k + n$
Rest of the world as a whole	n